

What is LRU cache ?

- Stand for Least recently used
- It is a caching mechanism that evicts the least recently used item when the cache reaches its capacity.

How does it work ?

- Hash map + Doubly Linked List
- Most recently used at the head, Least recently used at the tail
- When use an item, find it by hash map, move it to the head
- When out of space → new item need to be added → remove least used item → remove the tail

Why is a doubly linked list preferred over a singly linked list in an LRU cache implementation?

- A doubly linked list allows for constant-time removal and insertion of nodes from anywhere in the list by maintaining both previous and next pointers, which is essential for efficiently updating the cache's usage order.